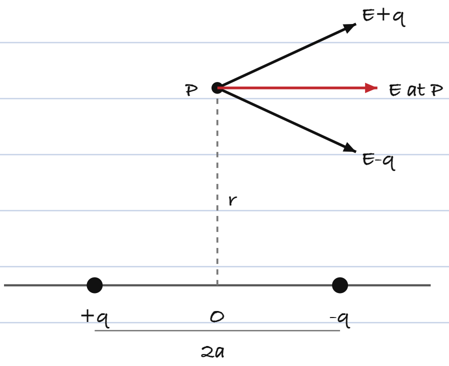
ANSWER BOOK

Physics (042) - Class XII Board Examination 2026 - Q.P. Code 55/1/1

Name: Ananya Sharma Roll No: 7261084 Section E only

Q31. (a)



The dipole has charges +q and -q separated by 2a. P is a point on the equatorial plane at a distance r from the centre O.

Distance of P from each charge = sqrt(r^2 + a^2)

E+q = (1/4 pi e0) . q/(r^2 + a^2) , along the line from +q to P

E-q = (1/4 pi e0) . q/(r^2 + a^2) , along the line from P to -q

Both are equal in magnitude.

The components perpendicular to the axis cancel out and the components parallel to the axis add up.

So E = (E+q + E-q) cos(theta) where cos(theta) = a / sqrt(r^2 + a^2)

E = 2 x (1/4 pi e0) x q/(r^2 + a^2) x a/sqrt(r^2 + a^2)

E = (1/4 pi e0) . 2qa / (r^2 + a^2)^(3/2)

For a far off point r >> a, so (r^2 + a^2)^(3/2) is nearly r^3

E = (1/4 pi e0) . 2qa/r^3 = p / (4 pi e0 r^3)

(b)

Charge +q is at x = a and charge -q is at x = b, and E = 2 i N/C.

Force on +q = qE = 2q i

Force on -q = -qE = -2q i

Net force F = 2q i - 2q i = 0 N

(the field is uniform so the two forces are equal and opposite)

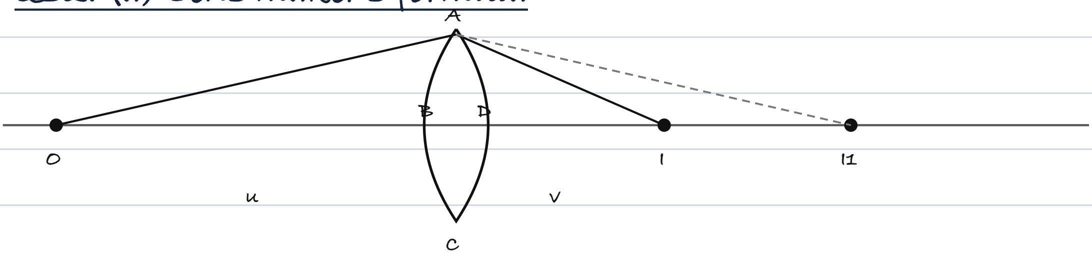
Torque, t = p E sin(theta)

The dipole is placed in the x-y plane and E is along x, so I take the angle between p and E as theta = 90 degrees.

$p = q \times (b - a)$

$\underline{t = q(b - a) \times 2 \times \sin 90 = 2q(b - a)\ \text{N m}}$

Q32. (a) Lens maker's formula



Let a thin lens of refractive index n2 be kept in a medium of refractive index n1. The two surfaces have radii of curvature R1 and R2.

The first surface ABC forms the image of the object O at I1.

$n1/OB + n2/BI1 = (n2 - n1)/BC1 \qquad \ldots(1)$

For the second surface ADC, I1 acts as a virtual object and the final image is formed at I.

$-n2/DI1 + n1/DI = (n2 - n1)/DC2 \qquad \ldots(2)$

For a thin lens BI1 = DI1.

Adding (1) and (2),

$n1/OB + n1/DI = (n2 - n1)\ [1/BC1 + 1/DC2]$

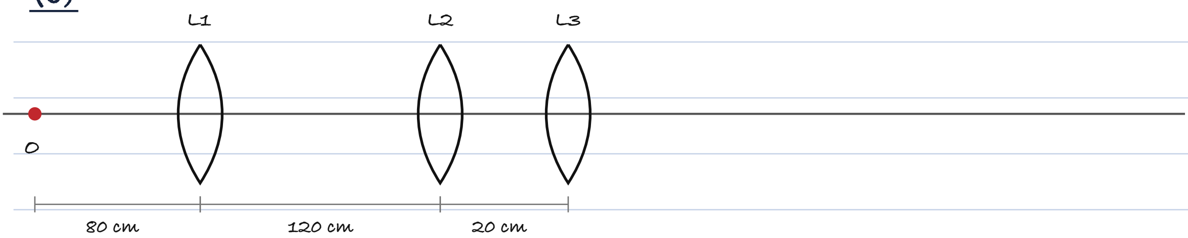
$-n1/u + n1/v = (n2 - n1)\ [1/R1 - 1/R2]$

$1/v - 1/u = (n2/n1 - 1)\ [1/R1 - 1/R2]$

$\underline{\text{So } 1/f = (n2/n1 - 1)\ [1/R1 - 1/R2]}$

which is the lens maker's formula.

(b)



$f_1 = f_2 = f_3 = 40$ cm and $u_1 = -80$ cm

For L1 : $1/v_1 - 1/(-80) = 1/40$

$1/v_1 = 1/40 - 1/80 = 1/80$

$v_1 = 80$ cm (to the right of L1)

For L2 : L1 L2 = 120 cm, so $u_2 = -(120 - 80) = -40$ cm

$1/v_2 = 1/40 + 1/(-40) = 0$

v_2 = infinity (parallel beam leaves L2)

For L3 : u_3 = infinity

$1/v_3 = 1/40$, so $v_3 = 40$ cm

The final image is 40 cm to the right of L3.

Distance of final image from the object = 120 + 20 + 40

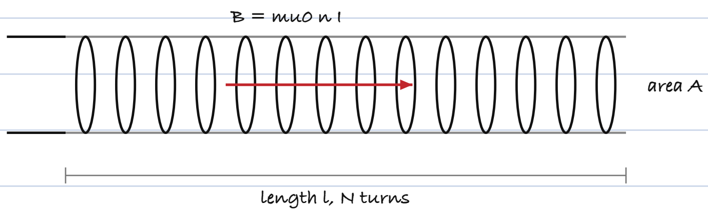
= 180 cm

Q33. (a)

Faraday's law : whenever the magnetic flux linked with a closed circuit changes, an e.m.f. is induced in it. The magnitude of the induced e.m.f. is equal to the rate of change of magnetic flux linked with the circuit.

$e = - d(\phi_B)/dt$

(b)



Let the solenoid have length l, area of cross section A and N turns, so the number of turns per unit length is $n = N/l$.

Magnetic field inside a long solenoid, $B = \mu_0 n I$

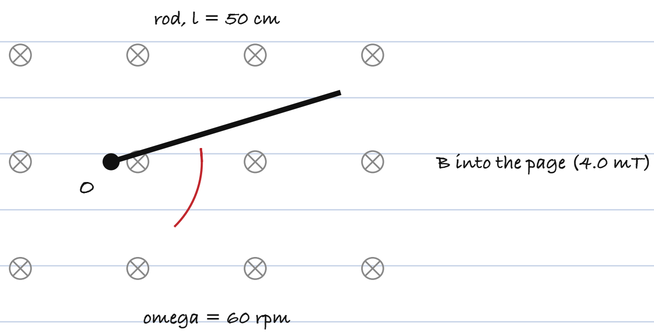
Flux through one turn $= B A = \mu_0 n I A$

Total flux linked, $N \phi_B = (n l)(\mu_0 n I)(A) = \mu_0 n^2 A l I$

Self inductance $L = N \phi_B / I$

$\underline{L = \mu_0 n^2 A l = \mu_0 N^2 A / l}$

(c)



$l = 50\ \text{cm} = 0.5\ \text{m}$, $B = 4.0\ \text{mT} = 4 \times 10^{-3}\ \text{T}$

$\omega = 60\ \text{rpm} = 60\ \text{rad/s}$

For a rod pivoted at one end,

$e = (1/2) B l^2 \omega$

$= (1/2) \times 4 \times 10^{-3} \times (0.5)^2 \times 60$

$= (1/2) \times 4 \times 10^{-3} \times 0.25 \times 60$

$\underline{= 0.03\ \text{V} = 30\ \text{mV}}$